pfi

Python for Fixed Income

COURSE 7 · NLP, RAG, AND LLMS

Clustering 10-K Risk Factors

WEEK 2 · DAY 3

Created by **Jeff Lenamon**

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From search to grouping

Day 6: a question, then a search

The question chose what came back: semantic search over an indenture.

Today: no question, no label

K-means groups the risk factors; the course describes each group in the filers' own words.

18

risk factors, Boyd Gaming 10-K

48

risk factors, Levi Strauss 10-K

33

risk factors, Lamar 10-K

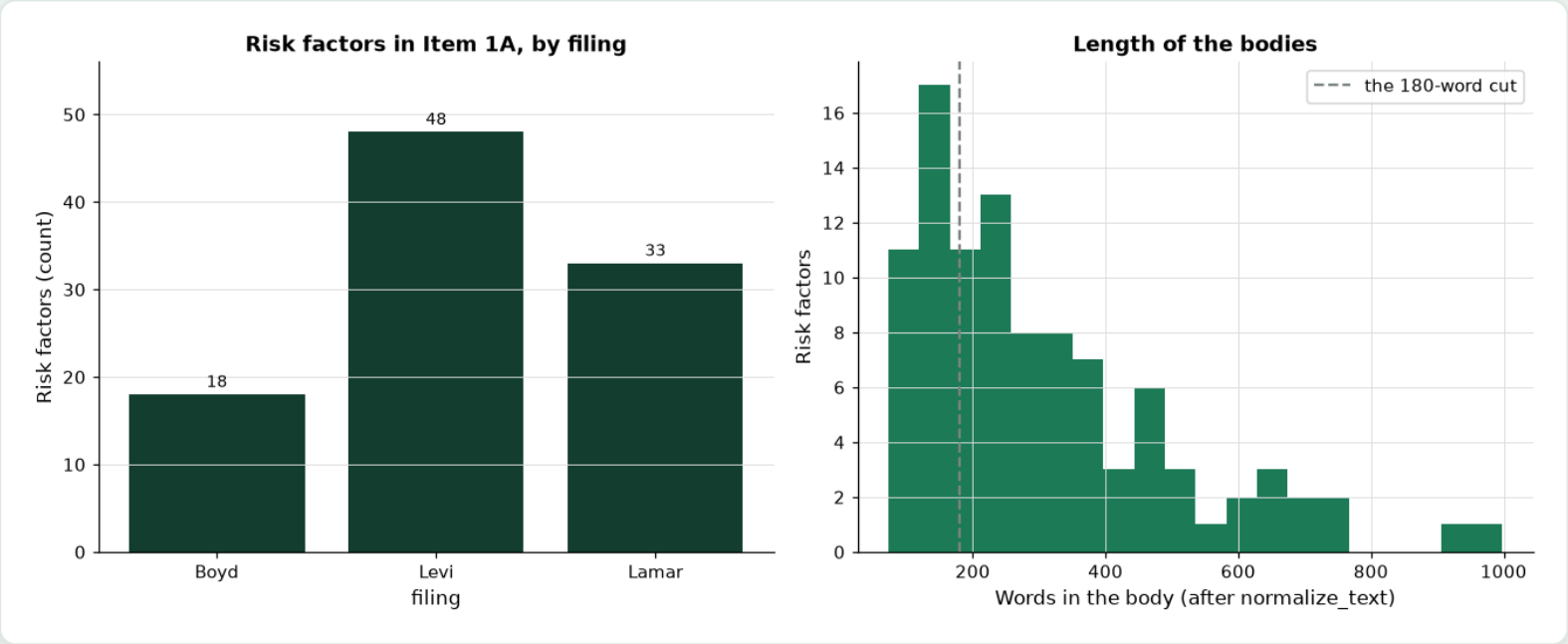
Clusters, described

A cluster of 10-K risk factors: its number, its terms, its counts per filing

Allowed names	Never called
a cluster	a risk ranking of the issuers
the risk factors in cluster 3	a measure of exposure
count of risk factors per filing	a credit view on a filer

A count of paragraphs is not a measure of risk: it shows how a filer drafts Item 1A.

Risk factors, counted per filing



99 risk factors; a body runs 73 to 998 words, median 241; 67 are longer than 180.

The counts describe the documents. They never rank the filers.

Source: Item 1A of the three Form 10-K filings, fiscal 2025, course7_risk_factors.csv.

One embedding per risk factor

```
texts = [heading + " " + first 180 words of the body]   # normalized
matrix = textkit.embed(model, texts)                  # once; every row of length 1
```

244

word pieces, the median text

26

of 99 texts over 256 pieces

1.4481

squared distance = 2 - 2 x cosine

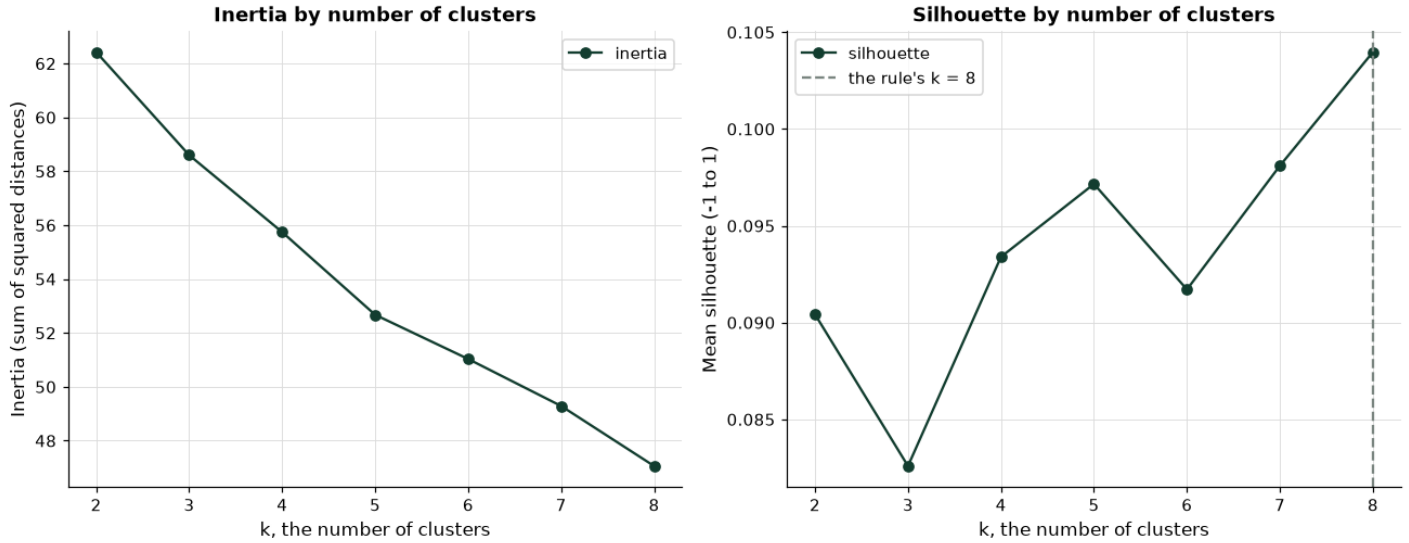
On unit-length rows, K-means groups by direction: the cosine of day 6.

How K-means and the silhouette work

The problem	Group rows with no labels, around k centres
The loop	Assign each row to its nearest centre, move each centre to the mean of its rows, repeat
Ten starts	The result depends on the first centres: keep the run with the lowest inertia
Silhouette	How much nearer a row is to its own cluster than to the next, -1 to 1; the mean scores k
What comes out	A label for each row, the centres, and the inertia

K-means, and the rule for k

The rule, stated first: k from 2 to 8, the highest silhouette.



8

clusters, by the rule

0.1040

silhouette: clusters overlap

Every silhouette is near 0.1: a way to sort the risk factors for reading, not groups that stand far apart.

How a cluster is described

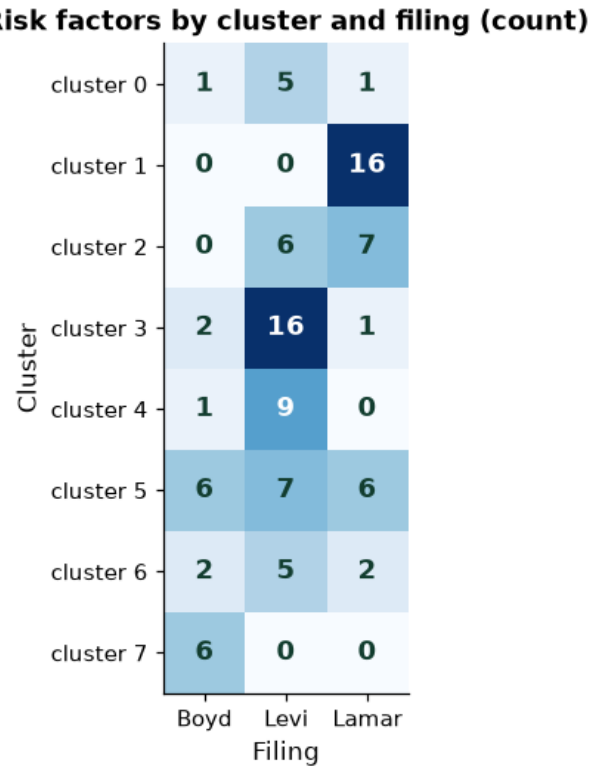
The problem	A cluster is row numbers; say what its texts share, in words
TF-IDF	Fitted on all the texts, single words, English stop words out
Lift	A term's mean weight inside the cluster minus its mean outside
Ties	Equal lifts go to the term that comes first alphabetically
What comes out	Each cluster's size and terms: a description, never a name

Describing a cluster: its terms

Cluster	Size	First five terms (the filers' words)
0	7	class, stock, common, securities, price
1	16	lamar, reit, income, advertising, taxable
2	13	outdoor, advertising, products, marketing, displays
3	19	retail, natural, distribution, stores, customers
4	10	laws, sanctions, countries, regulations, trade
5	19	debt, credit, facility, cash, indebtedness
6	9	information, data, systems, security, personal
7	6	gaming, casino, casinos, facilities, local

Lift: a term's mean TF-IDF inside the cluster minus outside it. Called by number, never named.

Clusters by filing, as counts



Cluster	Boyd	Levi	Lamar	All
0	1	5	1	7
1	0	0	16	16
2	0	6	7	13
3	2	16	1	19
4	1	9	0	10
5	6	7	6	19
6	2	5	2	9
7	6	0	0	6

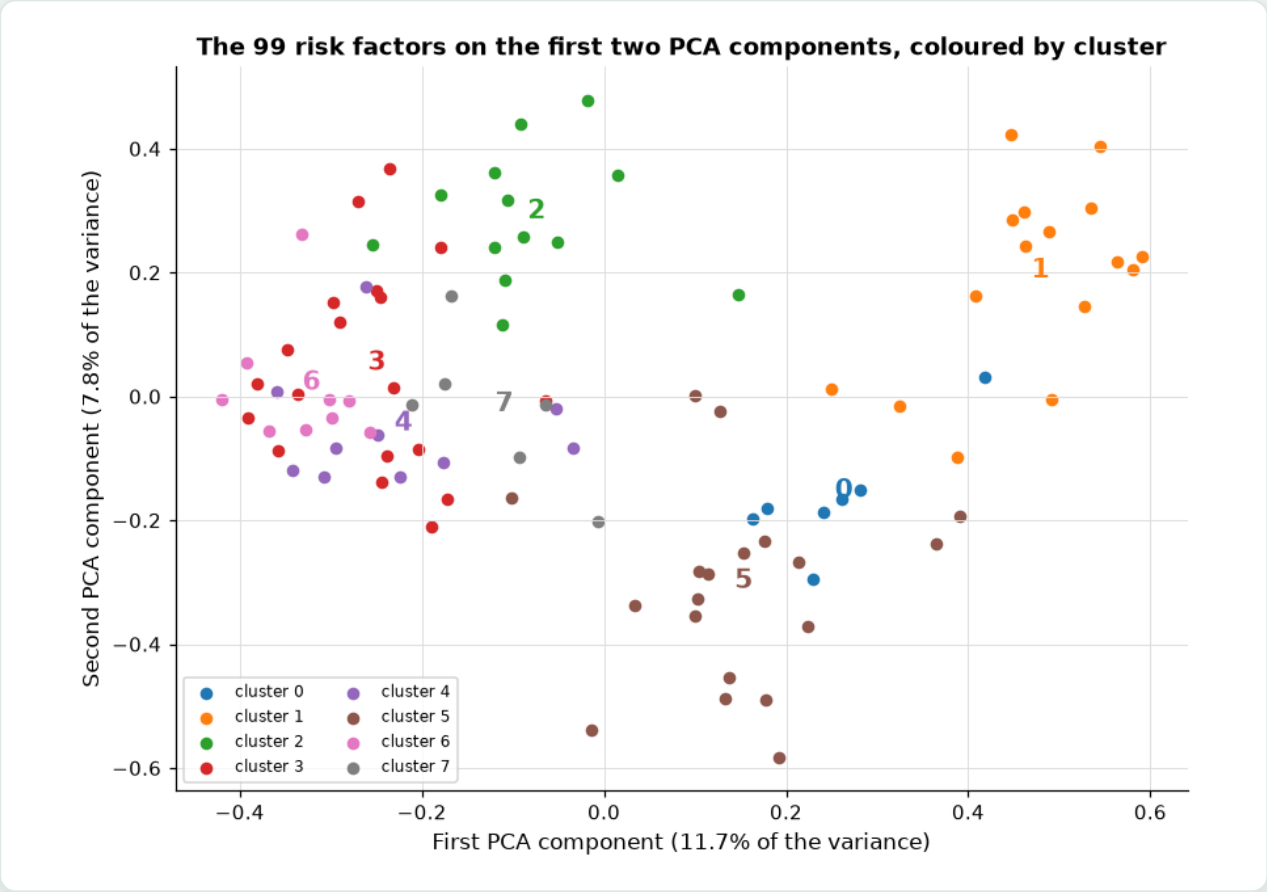
A count of paragraphs is not a measure of risk.

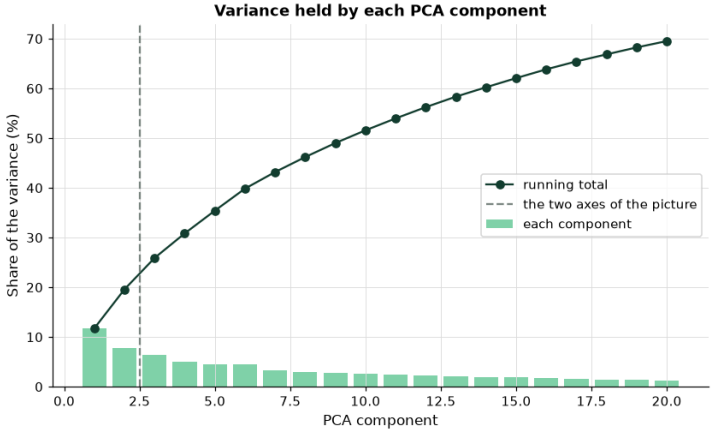
The clusters move with the model, the 180-word cut, and k.

How PCA draws it

The problem	384 numbers per row cannot be drawn
Directions	PCA finds the directions along which the rows spread most, in order
Two axes	The first two keep the biggest part of the spread two numbers can hold
What you control	The number of components
The catch	Rows together on the picture can be far apart in all 384 numbers

A picture in two dimensions, and what it hides





0.1958

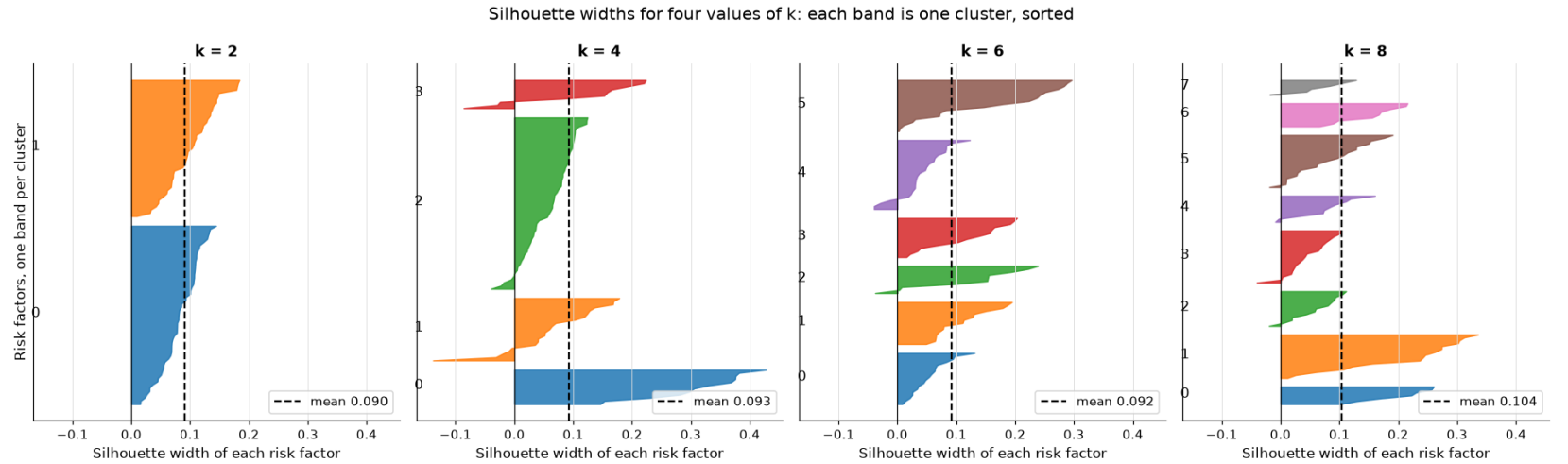
of the variance on the two axes

0.8042

left off the picture

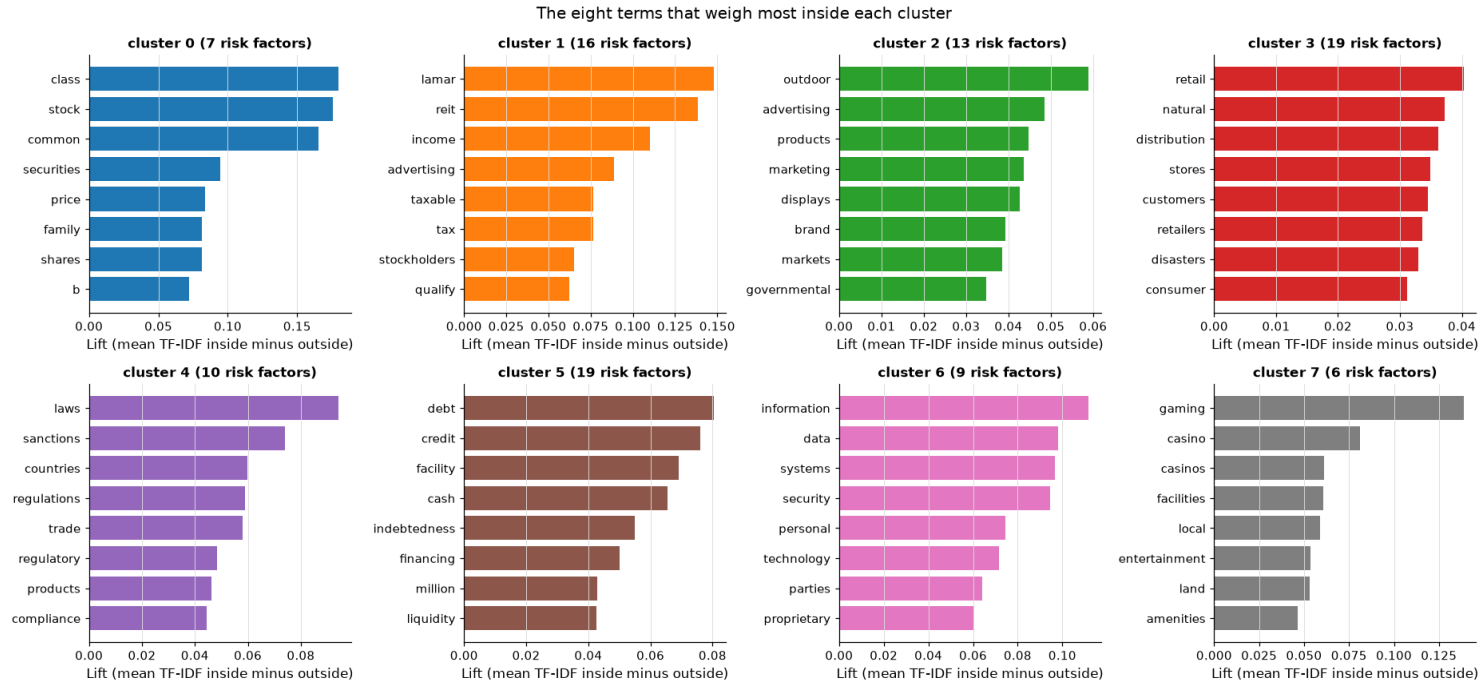
10 components hold half the spread, 48 hold 90 percent. A sketch, never a test.

Silhouette diagrams



k	Mean silhouette	Negative widths	Smallest cluster
2	0.0904	0	43
4	0.0934	13	10
6	0.0917	5	10
8	0.1040	6	6

Each cluster's words



A steep first bar: one word describes the cluster. Flat bars: a loosely shared vocabulary. Terms describe, they never name.

How the settings of KMeans work

The problem	Nine settings: which ones change the clusters?
How many	n_clusters: more always lowers inertia, so choose by silhouette
Where it starts	init and n_init: first centres, and how many runs from other ones
When it stops	max_iter and tol: limits a run seldom reaches
Always set	random_state: another seed is another answer

Every setting of KMeans

n_clusters (default 8; course "the rule's k")

How many clusters.

init (default 'k-means++'; course default)

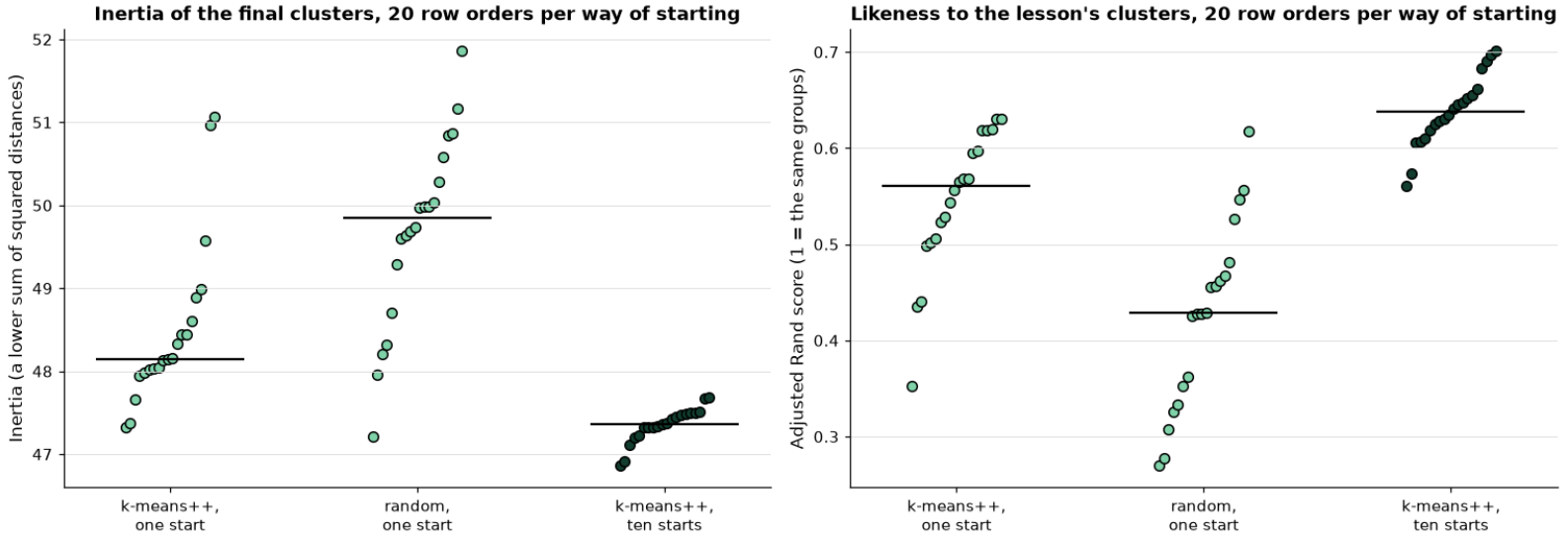
How the first centres are chosen: 'k-means++' spreads them out, 'random' picks rows at random.

n_init (default 'auto'; course 10)

How many times the whole fit runs from different first centres; the run with the lowest inertia is kept. 'auto': 1 for k-means++, 10 for random.

9 settings, read from the installed library; the three shaded are the ones to tune first. The course sets 3 and leaves the rest at their defaults.

What the starting settings do



Way of starting	Lowest	Highest	Orders at the lowest	Mean Rand score
k-means++, one start	47.320	51.072	0 of 20	0.55
random, one start	47.213	51.868	0 of 20	0.43
k-means++, ten starts	46.858	47.686	1 of 20	0.64

Excel side by side

Excel side by side (tested in Excel)

`=COUNTIFS(risk_factors!$A$2:$A$100,B$1,risk_factors!$C$2:$C$100,$A2)`	grid equal: True
`=COUNTIFS(risk_factors!A2:A100,"Levi Strauss & Co.",risk_factors!C2:C100,3)`	16
`=COUNTIF(risk_factors!$A$2:$A$100,B$1)`	18, 48, 33
`=SUM(B2:D9)`	99

No sentence model or K-means in Excel: the clusters come from Python, the counting is COUNTIFS.

Save the day: one file's patch

```
git log -p -1 -- textkit.py
+def cluster_terms(texts: list[str], labels, n: int = 8) -> pd.DataFrame:
```

28 added lines, nothing removed. pytest: 42 passed.

The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.

1

Docstring

Units, conventions, and what the function must never do: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Assert what the docstring promises. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: the piece vectors are summed, never scaled to length 1: rows 1.0000 to 5.0102, clusters by length.

Code that runs is not the same as code that is right.

What you can do now

- Count risk factors per filing, with source lines
- Embed once; unit rows make distance follow cosine
- Choose k by silhouette; know the starting settings
- Describe a cluster by its terms, never a name
- Counts by filing: crosstab and COUNTIFS

Week 2, Day 4 Reading EDGAR